COURSE 2 · ALGEBRA · MODULE 2.3

Rearranging Formulas

Visual companion and Symbols Sheet

Page 1 is the Symbols Sheet: every numbered equation from the audio reader. Pages 2 to 5 are the diagrams: the balance, the build tree, the traps, linearisation, and the seven-step procedure as a flowchart.

The animated explainer video covers the same diagrams in motion.

Page 1

Symbols Sheet

Every numbered equation from the audio theory reader, in conventional notation. Work down the list with a pencil and, for each one, say the spoken version aloud before reading the symbols.

1	First equation of motion	$v = u + at$
2	...rearranged for acceleration	$a = \frac{v-u}{t}$
3	Simple pendulum	$T = 2\pi\sqrt{\frac{L}{g}}$
4	...rearranged for g	$g = \frac{4\pi^2 L}{T^2}$
5	...rearranged for L	$L = \frac{gT^2}{4\pi^2}$
6	Thin lens	$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$
7	...rearranged for v	$v = \frac{uf}{u-f}$
8	Rational function and its inverse	$y = \frac{x+2}{x-3} \iff x = \frac{3y+2}{y-1}$
9	Radioactive decay	$N = N_0 e^{-kt}$
10	...rearranged for time	$t = \frac{\ln(N_0/N)}{k}$
11	Compound growth, for time	$A = P(1+r)^t \Rightarrow t = \frac{\ln(A/P)}{\ln(1+r)}$
12	pH and concentration	$\mathrm{pH} = -\log_{10}[\mathrm{H}^+] \iff [\mathrm{H}^+] = 10^{-\mathrm{pH}}$
13	Pendulum, linearised	$T^2 = \frac{4\pi^2}{g} L$
14	Decay, linearised	$\ln N = \ln N_0 - kt$
15	Power law, linearised	$\log y = \log a + n \log x$
16	Kinetic energy, for speed	$E = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{\frac{2E}{m}}$

The undo list

Operation	Undone by	Watch out for
add a	subtract a	apply to the whole side
multiply by m	divide by m	m must not be zero
square, x^n	square root, x^(1/n)	± for even powers
e^x or 10^x	ln x or $\log_{10}$ x	logs need positive inputs
sin x	arcsin x	more than one angle may fit
reciprocal 1/x	reciprocal again	only of a whole side

Page 2 · Reader sections 2 and 4

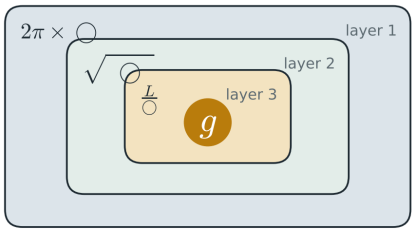
The balance and the build tree

Start Subtract u, both sides Divide by t, both sides



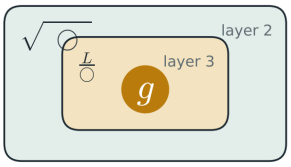
Whatever you do to one pan you do to the other. Moving terms 'across the equals sign' is shorthand for this, and the shorthand fails precisely when you forget it.

$$T = 2\pi\sqrt{\frac{L}{g}}$$



Step 0: Three layers wrap g. Peel from the outside in.

$$\frac{T}{2\pi} = \sqrt{\frac{L}{g}}$$



Step 1: Undo × 2π : divide both sides by 2π

$$\frac{T^2}{4\pi^2} = \frac{L}{g}$$



Step 2: Undo √ : square both sides

$$g = \frac{4\pi^2 L}{T^2}$$



Step 3: Undo L ÷ g : take reciprocals, multiply by L

The pendulum law, Equation 3, rearranged for g. Each box is one operation wrapped around g. The outermost box was applied last, so it is undone first: socks on, shoes on; shoes off, socks off.

Page 3 · Reader sections 7 and 8

Two traps and one routine

The reciprocal trap

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v} \quad /\Rightarrow \quad f = u + v$$

Test: one half plus one half is one, but two plus two is not one. The fix is to make each side a single fraction first:

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{u - f}{uf} \quad \Rightarrow \quad v = \frac{uf}{u - f}$$

The same rule governs resistors in parallel and springs in series.

Collect and factor: target appears twice

$$y = \frac{x + 2}{x - 3}$$

1	Clear the fraction	$y(x - 3) = x + 2$
2	Expand	$xy - 3y = x + 2$
3	Collect x-terms left	$xy - x = 3y + 2$
4	Factor out x	$x(y - 1) = 3y + 2$
5	Divide ($y \neq 1$)	$x = \frac{3y + 2}{y - 1}$

Squaring loses the sign

$$x^2 = 9 \;\Rightarrow\; x = \pm 3 \qquad E = \tfrac{1}{2}mv^2 \;\Rightarrow\; v = +\sqrt{\tfrac{2E}{m}} \quad (\text{speed} \geq 0)$$

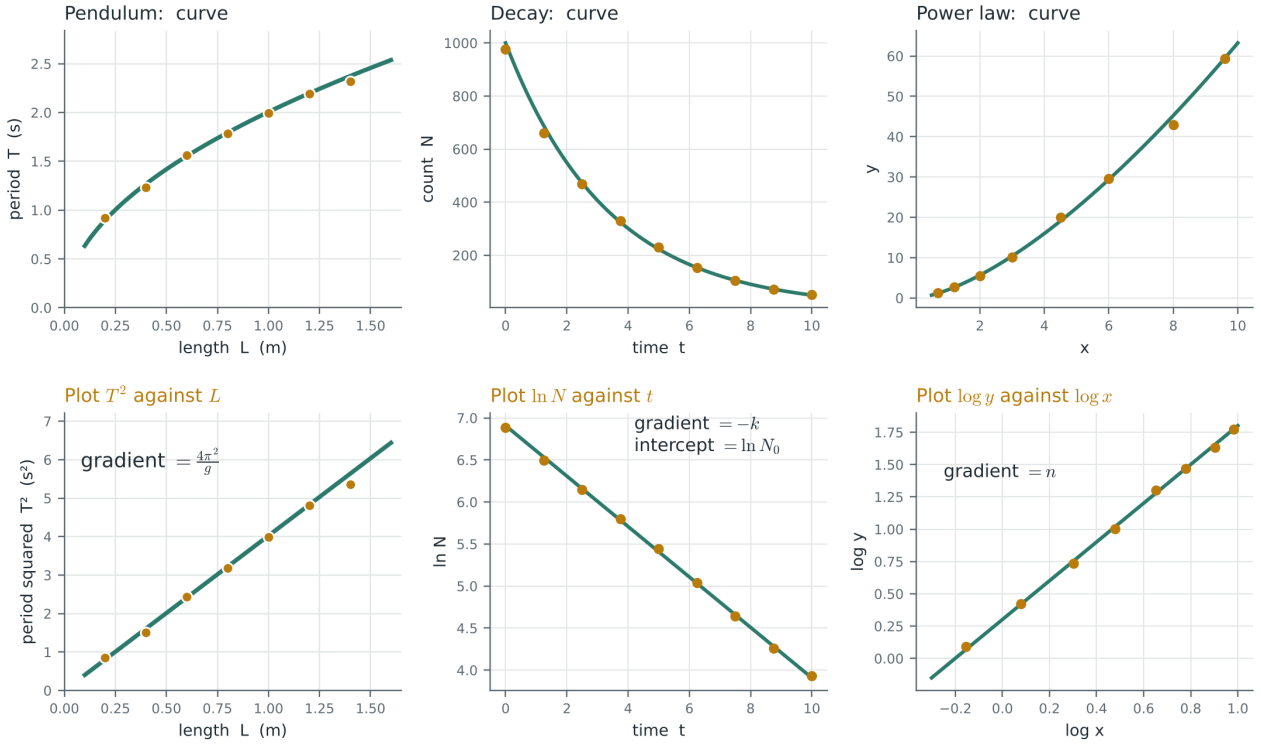
Four checks

Substitute	Pick easy numbers, compute the target, feed everything back into the original equation.
Units	The answer's units must be the target's units: acceleration in m/s^2, not m/s.
Limiting case	Send one input to zero or infinity. Lens: as u grows huge, v approaches f.
Scaling	Double an input; does the output change by the factor the formula predicts?

Page 4 · Reader section 12

Linearisation: turning curves into lines

Top row: the raw relationship. Bottom row: the rearranged or transformed version that plots as a straight line. Constants are read from the gradient and intercept. Equations 13, 14 and 15.

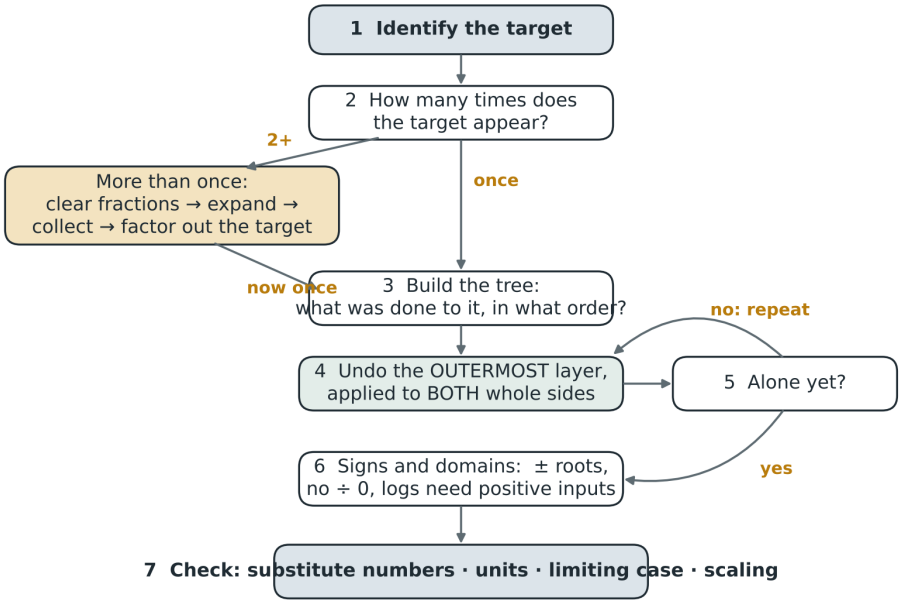


Bridge from the social sciences

The bottom-right panel is the log–log model behind a constant-elasticity regression, where the slope coefficient is the elasticity. The bottom-middle panel is the log-linear model, where the slope is a constant proportional growth or decay rate. You have run both; physics simply names the slope a physical constant.

Page 5 · Reader section 13

The procedure, as a flowchart



Keep this page beside you for the workbook. By the end of Set D you should no longer need it.